AN ASCENDING CHAIN CONDITION ON A GRADED PIECE FORCES THE MITTAG-LEFFLER CONDITION FOR THE COHOMOLOGY OF AN INVERSE SYSTEM OF SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument, including the graded compatibility diagram and the final stable-image computation; the author's closing sentence is retained verbatim.

Editorial note. The stated lemma carries a footnote spelling out the stable-image form of Mittag-Leffler; the footnote lies inside the counted statement span and is delivered as the author wrote it.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology (topological spaces) and in chapter more-algebra. The conclusion is the same and those statements are special cases of this site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Mittag-Leffler condition for an inverse system of modules. Same chapter, the Mittag-Leffler condition as the stabilisation of the images Im(M_n -> M_{n-c}). Supplies the conclusion, in the equivalent stable-image form stated in the lemma's footnote.
- Long exact cohomology sequence of a short exact sequence of sheaves. Same chapter, the boundary map delta_n : M_n -> N_n attached to 0 -> I^n F_{n+1} -> F_{n+1} -> F_n -> 0. Supplies the boundary maps whose images are shown to form a finitely generated graded module.
- Graded module structure over the Rees-type ring of the powers of I. Same chapter, the graded module structure of the direct sum of the N_n over the direct sum of the I^n/I^{n+1}. Supplies the ascending chain condition hypothesis, which is applied to the graded submodule of the images of the delta_n.
- Commutative diagram relating the boundary maps for n and n + m. Same chapter, the morphism of short exact sequences induced by multiplication by an element f of I^m, giving delta_{n+m}(fs) = f delta_n(s). Supplies the compatibility that makes the direct sum of the images of the delta_n a graded submodule.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let I be an ideal of a ring A. Let $\mathcal{C}$ be a site. Let*

$$\ldots \to \mathcal{F}_3 \to \mathcal{F}_2 \to \mathcal{F}_1$$

be an inverse system of sheaves of A-modules on $\mathcal{C}$ such that $\mathcal{F}_n = \mathcal{F}_{n+1}/I^n\mathcal{F}_{n+1}$. Let $p \geq 0$. Assume

$$\bigoplus\nolimits_{n\geq 0} H^{p+1}(\mathcal{C}, I^n\mathcal{F}_{n+1})$$

satisfies the ascending chain condition as a graded $\bigoplus_{n\geq 0} I^n/I^{n+1}$-module. Then the inverse system $M_n = H^p(\mathcal{C}, \mathcal{F}_n)$ satisfies the Mittag-Leffler condition[1].

Proof. Set $N_n = H^{p+1}(\mathcal{C}, I^n\mathcal{F}_{n+1})$ and let $\delta_n : M_n \to N_n$ be the boundary map on cohomology coming from the short exact sequence $0 \to I^n\mathcal{F}_{n+1} \to \mathcal{F}_{n+1} \to \mathcal{F}_n \to 0$. Then $\bigoplus \mathrm{Im}(\delta_n) \subset \bigoplus N_n$ is a graded submodule. Namely, if $s \in M_n$ and $f \in I^m$, then we have a commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & I^n\mathcal{F}_{n+1} & \longrightarrow & \mathcal{F}_{n+1} & \longrightarrow & \mathcal{F}_n & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f} & & \\
0 & \longrightarrow & I^{n+m}\mathcal{F}_{n+m+1} & \longrightarrow & \mathcal{F}_{n+m+1} & \longrightarrow & \mathcal{F}_{n+m} & \longrightarrow & 0
\end{array}$$

The middle vertical map is given by lifting a local section of $\mathcal{F}_{n+1}$ to a section of $\mathcal{F}_{n+m+1}$ and then multiplying by f; similarly for the other vertical arrows. We conclude that $\delta_{n+m}(fs) = f\delta_n(s)$. By assumption we can find $s_j \in M_{n_j}$, $j = 1, \ldots, N$ such that $\delta_{n_j}(s_j)$ generate $\bigoplus \mathrm{Im}(\delta_n)$ as a graded module. Let $n > c = \max(n_j)$. Let $s \in M_n$. Then we can find $f_j \in I^{n-n_j}$ such that $\delta_n(s) = \sum f_j\delta_{n_j}(s_j)$. We conclude that $\delta(s - \sum f_j s_j) = 0$, i.e., we can find $s' \in M_{n+1}$ mapping to $s - \sum f_j s_j$ in M_n. It follows that

$$\mathrm{Im}(M_{n+1} \to M_{n-c}) = \mathrm{Im}(M_n \to M_{n-c})$$

Namely, the elements $f_j s_j$ map to zero in M_{n-c}. This proves the lemma. □

[1] In fact, there exists a $c \geq 0$ such that $\mathrm{Im}(M_n \to M_{n-c})$ is the stable image for all $n \geq c$.